

Hailstone Numbers

& Collatz Conjecture



What are Hailstone Numbers?

The Hailstone sequence is simple:

$$\begin{aligned} X_{n+1} &= \frac{1}{2} \cdot X_n & ; & & X_n \% 2 == 0 \text{ (Even)} \\ X_{n+1} &= 3 \cdot X_n + 1 & ; & & X_n \% 2 == 1 \text{ (Odd)} \end{aligned}$$

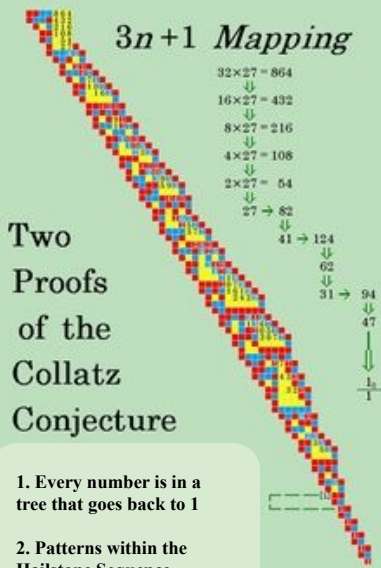
Where does this sequence end?

*This sequence never ends.
Instead, it enters an infinite loop.*

$$4 \rightarrow 2 \rightarrow 1$$

$$\begin{aligned} 4 / 2 &= 2 \\ 2 / 2 &= 1 \\ 3(1) + 1 &= 4 \end{aligned}$$

Hiroiyuki Kawasaki



Collatz Conjecture:

The conjecture asks whether repeating two simple arithmetic operations will eventually transform every positive integer into 1

So far, it has been proven true for numbers up to $2.36 \cdot 10^{21}$

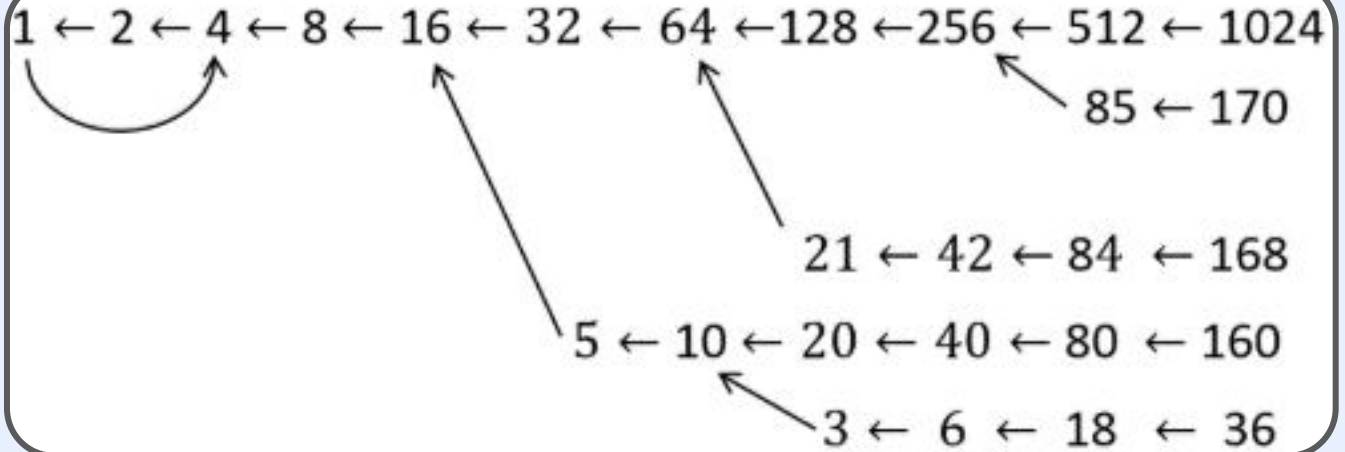
Visualization & Calculator

The sequence on the top right is shows how various numbers always end up in the pattern of $4 \rightarrow 2 \rightarrow 1$.

The python program on the bottom right is a calculator to print the entire sequence of any number until it reaches 1.

A demo of the calculator will be shown.

% = Remainder Of



```
val = int(input("Enter number: "))
while True:
    print(val)
    if val == 1: break
    elif val % 2 == 0: val = int(val/2)
    else: val = int(3 * val + 1)
```

Thank you

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